where we applied the fixed point Proposition A.6. By Assumption A.10, the left summand in (45) is at most $\varepsilon_{\text{OPT}}+\frac{a}{\sqrt{n}}\sqrt{\log\frac{b}{\delta}}$, with probability $1-\delta$. Next, recall that in (42) in the proof of Theorem A.8, we proved that that $\pi_\gamma G_\lambda^\infty$ is a contraction mapping of factor $\rho(\lambda)/(1+\gamma)$ sufficiently close to U; this implies that the right summand in (45) is at most $\frac{\rho(\lambda)}{1+\gamma}\|\theta_{t-1}^n-\theta^\star\|$. Together, these yield the recurrence estimate

$$\mathbb{P}\left(\|\theta_t^n-\theta^\star\|\leq\varepsilon_{\text{OPT}}+\frac{a}{\sqrt{n}}\sqrt{\log\frac{b}{\delta}}+\frac{\rho(\lambda)}{1+\gamma}\|\theta_{t-1}^n-\theta^\star\|\right)\geq 1-\delta. \tag{46}$$

Iterating this recurrence for successive time steps yields

$$\mathbb{P}\left(\|\theta_t^n-\theta^\star\|\leq\left(\varepsilon_{\text{OPT}}+\frac{a}{\sqrt{n}}\sqrt{\log\frac{b}{\delta}}\right)\sum_{i=0}^{t}\left(\frac{\rho(\lambda)}{1+\gamma}\right)^i+\left(\frac{\rho(\lambda)}{1+\gamma}\right)^t\|\theta_0^n-\theta^\star\|\right)\geq(1-\delta)^t. \tag{47}$$

Note that (47) holds for any $\delta\in(0,1)$. In particular, we can apply (47) with $\delta:=\delta/t$. In this case, the Bernoulli inequality lets us estimate that $(1-\delta/t)^t\geq 1-\delta$. This completes the proof, with $\delta_2=\delta$. □

Remark A.12. Theorem A.11 recovers the result from (Bertrand et al., 2024) in the case where the correction strength is $\gamma=0$. But for a fixed augmentation percentage λ, for any correction strength $\gamma>0$, this gives stronger stability guarantees than in (Bertrand et al., 2024).

Remark A.13. In a previous version of this manuscript, we claimed that there was an error in the statement of the corresponding theorem in (Bertrand et al., 2024). In this version, we retract that claim; we have corresponded with those authors, and they updated their manuscript with additional details to justify their statement.

A.4. Discussion: The Main Limitation

Our empirical results are for generative modeling tasks where we have access to some "self-correction" operation that is easy to compute, as well as automatic; see Sections 6 and 7 for more details about these correction functions. Therefore, the main limitation of our work is that one can only hope to use this self-correction procedure to stabilize training in scenarios where there is some "self-correction" function. For our MNIST experiments, we built a self-correction function from scratch using clustering statistics. And for our human motion experiments, we used an off-the-shelf human motion imitation model that other researchers built.

B. Discussion about Assumption 4.2

In this section, we show how with a mild boundedness assumption on our generative model parameter update function, we can deduce our Assumption A.10 (which is the same as Assumption 4.2, part 3) from the following assumption used in (Bertrand et al., 2024).

Assumption B.1. There exist $a,b,\varepsilon_{\text{OPT}}\geq 0$ and a neighborhood U of $\theta^\star$ such that, for any $\delta\in(0,1)$, with probability $1-\delta$ over the samplings, we have

$$(\forall\theta\in U)(\forall n\in\mathbb{N})\qquad\|\mathcal{G}_\lambda^n(\theta)-\mathcal{G}_\lambda^\infty(\theta)\|\leq\varepsilon_{\text{OPT}}+\frac{a}{\sqrt{n}}\sqrt{\log\frac{b}{\delta}}. \tag{48}$$

Now, if we make the additional assumption that our generative model parameter update function is locally bounded near $\theta^\star$ then we obtain the following.

Proposition B.2. *Suppose Assumption B.1 holds. Suppose also that there exists $B<\infty$ such that for all $n>0$ and θ sufficiently close to $\theta^\star$,*

$$\|\mathcal{G}_\lambda^n(\theta)-\mathcal{G}_\lambda^n(\theta^\star)\|<B\|\theta-\theta^\star\|.$$

Then there exist $a,b,c,\varepsilon_{OPT}\geq 0$ and a neighborhood U of $\theta^\star$ such that, for any $\delta\in(0,1)$, with probability $1-\delta$ over the samplings, we have

$$(\forall\theta\in U)(\forall n\in\mathbb{N})\qquad\|\pi_\gamma\mathcal{G}_\lambda^n(\theta)-\pi_\gamma\mathcal{G}_\lambda^\infty(\theta)\|\leq c\cdot d_U+\varepsilon_{OPT}+\frac{a}{\sqrt{n}}\sqrt{\log\frac{b}{\delta}}, \tag{49}$$

where $d_U=\sup_{\theta\in U}\|\theta-\theta^\star\|$.

Proof. By the triangle inequality, we have

$$\|\pi_\gamma \mathcal{G}_\lambda^n(\theta) - \pi_\gamma \mathcal{G}_\lambda^\infty(\theta)\| \leq \|\pi_\gamma \mathcal{G}_\lambda^n(\theta) - \mathcal{G}_\lambda^n(\theta)\| + \|\mathcal{G}_\lambda^n(\theta) - \mathcal{G}_\lambda^\infty(\theta)\| + \|\mathcal{G}_\lambda^\infty(\theta) - \pi_\gamma \mathcal{G}_\lambda^\infty(\theta)\|. \tag{50}$$

We bound each term in the RHS: firstly, note the middle term is bounded by Assumption B.1.The first term is bounded as follows:

$$\begin{aligned}\|\mathcal{G}_\lambda^n(\theta) - \pi_\gamma \mathcal{G}_\lambda^n(\theta)\| &\leq \|\mathcal{G}_\lambda^n(\theta) - \mathcal{G}_\lambda^n(\theta^\star)\| + \|\pi_\gamma \mathcal{G}_\lambda^n(\theta^\star) - \pi_\gamma \mathcal{G}_\lambda^n(\theta)\| \\ &\leq B\|\theta - \theta^\star\| + B\|\theta - \theta^\star\| \\ &\leq 2Bd_U,\end{aligned}$$

where in the first step we used that $\mathcal{G}_\lambda^\infty(\theta^\star) = \pi_\gamma \mathcal{G}_\lambda^\infty(\theta^\star)$. Similarly, the last term is bounded as follows:

$$\begin{aligned}\|\mathcal{G}_\lambda^\infty(\theta) - \pi_\gamma \mathcal{G}_\lambda^\infty(\theta)\| &\leq \|\mathcal{G}_\lambda^\infty(\theta) - \mathcal{G}_\lambda^\infty(\theta^\star)\| + \|\pi_\gamma \mathcal{G}_\lambda^\infty(\theta^\star) - \pi_\gamma \mathcal{G}_\lambda^\infty(\theta)\| \\ &\leq \rho(\lambda)\|\theta - \theta^\star\| + \frac{\rho(\lambda)}{1+\gamma}\|\theta - \theta^\star\| \\ &= \rho(\lambda)\frac{2+\gamma}{1+\gamma}\|\theta - \theta^\star\| \\ &\leq \rho(\lambda)\frac{2+\gamma}{1+\gamma}d_U,\end{aligned}$$

where in the second step we applied (42). Using these bounds in (50) and taking $c = 2B + \rho(\lambda)\frac{2+\gamma}{1+\gamma}$ completes the proof. □

Note that the constant $c \cdot d_U < c$ (for U sufficiently small) can really be viewed as a part of the optimization constant ε_{OPT} since it is controlled by the choice of generative model class.

C. Point-wise correction corresponds to distribution-wise correction

In this section we provide a sufficient condition under which you can associate a distribution-wise correction mapping (like the one we consider in the paper, π_γ) to a point-wise correction mapping (which is the one you are more likely to find in the wild).

Definition C.1. Let $X = \{x_1, \ldots, x_n\} \subset \mathbb{R}^m$ and define the *empirical cumulative distribution* function Φ_X by

$$\Phi_X(v) := \Phi_X(v; \{x_1, \ldots, x_n\}) := \frac{1}{n}\sum_{i=1}^n \chi_v(x_i),$$

where for $v \in \mathbb{R}^m$, $\chi_v : \mathbb{R}^m \to \{0, 1\}$ is the indicator function for the set $\prod_{i=1}^n (-\infty, v_i]$. For a continuous distribution, the cumulative distribution function is defined in the usual way.

Definition C.2. Suppose that we have a model p_θ and an arbitrary function $\Pi : \mathbb{R}^m \to \mathbb{R}^m$. Then we say that Π is a *valid point-wise correction function* for p_θ if there exists a $\gamma \in [0, \infty]$ such that

$$\lim_{n\to\infty}\left(\mathbb{E}_{X^n \sim p_\theta^n} \sup_{v\in\mathbb{R}^m} \|\Phi_{\Pi(X^n)}(v) - \Phi_{\pi_\gamma p_\theta}(v)\|\right) \to 0, \tag{51}$$

almost surely, where the expectation is over all samplings $X^n = \{x_1, \ldots, x_n\}$ of size n from p_θ.

Intuition C.3. This is saying that the CDFs for $\pi_\gamma p_\theta$ and $\Pi(X \sim p_\theta^n)$ are equal in expectation, for large enough n. This is one way of saying that $\pi_\gamma p_\theta$ and $\Pi(X \sim p_\theta^n)$, for large enough n, are nearly identical probability distributions.

Definition C.4. If the limit in (51) exists, then we define the *distribution-wise projection function* corresponding to Π to be

$$\pi_\gamma p_\theta = \frac{1}{1+\gamma}p_\theta + \frac{\gamma}{1+\gamma}p_{\theta^\star}, \tag{52}$$

and we define the *projection strength of the point-wise correction function* Π to be γ. Recall that $\pi_\gamma p_\theta = \frac{1}{1+\gamma}p_\theta + \frac{\gamma}{1+\gamma}p_{\theta^\star}$. So intuitively, (51) implies that the projection function Π maps samples from p_θ to a different space such that they look like they come from a combination of the original distribution p_θ and $p_{\theta^\star}$, at least at the level of CDFs.

Remark C.5. Such a γ, if it exists, is unique. Furthermore, if $p_\theta = p_{\theta^\star}$, then $\gamma = \infty$.

The limit condition in Definition C.2 is abstract, and can be hard to swallow. We present an example of a simple point-wise correction for the Gaussian toy example that we consider in Section 5, whose corresponding distribution-wise correction is exactly one would expect it to be–the weighted average of the corresponding Gaussians. Recall that we demonstrated empirically in Figure 2 that Theorem 4.3 holds for that example. The projection function is depicted in Figure 6.

Example C.6. Let $G_1(x)$ be the pdf of $\mathcal{N}(0, \sigma_1^2 I_d)$ (initial distribution, corresponds to θ) and $G_2(x)$ the pdf of $\mathcal{N}(0, \sigma_2^2 I_d)$ (target distribution, corresponds to $\theta^\star$). Given $x_1, \ldots, x_n \sim G_1$, we define Π^γ as follows: Fix any $\gamma \in \mathbb{R}_{\geq 0}$, and let $y_1, \ldots, y_n \sim (\hat{G}_1^{(n)}(x) + \gamma G_2(x))/(1+\gamma)$, where $\hat{G}_1^{(n)}$ is the PDF of the empirical distribution defined by $\{x_1, \ldots, x_n\}$; in practice we implement $\hat{G}_1^{(n)}$ as a histogram. Then choose a random $\sigma \in S_n$ (S_n = group of permutations on n symbols). Finally, we define $\Pi^\gamma(x_i) := y_{\sigma(i)}$ for $1 \leq i \leq n$.

Next, we define the projection set $\Pi X^{(n)} := \{\Pi^\gamma(x_i)\}_{1 \leq i \leq n}$, and define the PDF $\pi_\gamma \hat{G}_1^{(n)}(x) := \frac{1}{1+\gamma}\hat{G}_1^{(n)}(x) + \frac{\gamma}{1+\gamma}G_2(x)$, and let $\Phi_{\pi_\gamma \hat{G}_1^{(n)}}$ represent the cumulative distribution function of the Gaussian $\pi_\gamma \hat{G}_1^{(n)}$. Then, since $\Pi^\gamma(x_i) \sim \pi_\gamma \hat{G}_1^{(n)}$, we have by the uniform law of large numbers that

$$\lim_{n\to\infty} \left(\mathbb{E}_{\{x_i \sim G_1\}_{i=1}^n} \sup_{v \in \mathbb{R}^m} \left\|\Phi_{\Pi X^{(n)}}(v) - \Phi_{\pi_\gamma G_1}(v)\right\|\right) \to 0 \tag{53}$$

almost surely. Therefore Π^γ is a valid point-wise correction function, and its corresponding distribution-wise projection function is π_γ.

Remark C.7. In the example we considered in Section 5, we could have included a total distance traveled minimization condition, but here for this proof we don't even need to use that hypothesis. (In the proof, this would have corresponded to the additional assumption that we've chosen a $\sigma \in S_n$ such that $\sum_{i=1}^n \|x_i - y_{\sigma(i)}\|$ is minimized.) This implies that different point-wise correction functions can correspond to the same distribution-wise correction function.

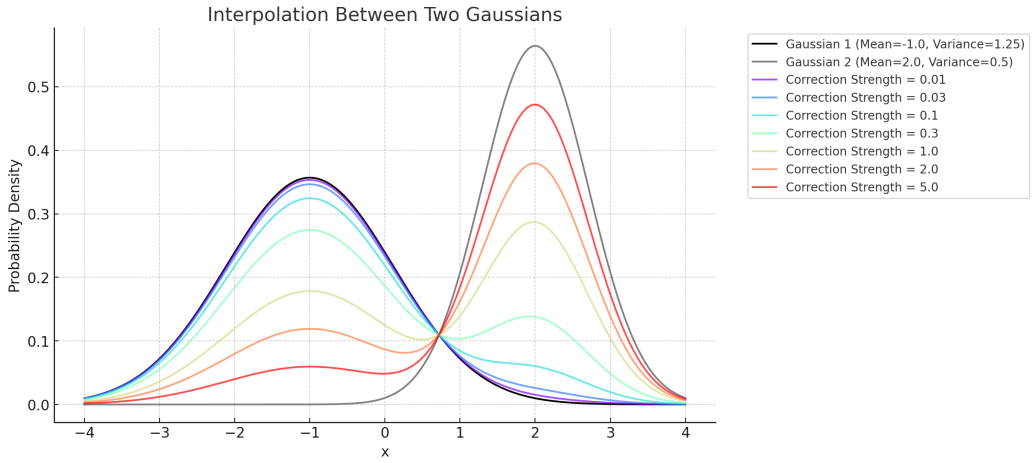

Figure 6. Illustration of the distribution-wise projection function, like in our Gaussian toy example. Correcting one Gaussian in the direction of another, like we consider in Section 5, corresponds to finding the "(weighted) average Gaussian" that lives between the two.

D. More MNIST Experiment Details

We train a Denoising Diffusion Probabilistic Model (DDPM) (Ho et al., 2020) on the 20% of the MNIST dataset (LeCun et al., 1998). We use classifier-free guidance (Ho & Salimans, 2021) with guidance parameter 0.5, and 400 diffusion steps. We used a batch size of 256. We train generation 0 for 20 epochs, with a linear decay learning rate schedule starting at $1e-4$ and ending at $(1e-4)/20$. We train each following generations for a single epoch, with a fixed learning rate of $(1e-4)/20^2$.

To compute our metrics, we first train a LeNet model (LeCun et al., 1998) on MNIST, and then we sample an equal number of digits from each class using the checkpoint that we're trying to evaluate. To compute the FID score, we extract embeddings